

1. Complete the factor trees and write the prime factorisation in index notation:

2. Use the Venn diagram (prime factorisation) method. Show all working:

3. True investigation: Is each number prime or composite? Show your divisibility checks:

4. Solve these word problems. State HCF or LCM, explain why, and show your working:

5. The HCF of two numbers is 12. The LCM of the same two numbers is 180. One of the numbers is 36. What is the other number? Show your working and explain your method.

6. A student writes: 'The prime factorisation of 36 is $2 \times 3 \times 6$.' Explain clearly why this is WRONG. Write the correct prime factorisation and use the lesson's method to show your working.

7. The lesson says we use base-60 mathematics every day, invented by the Mesopotamians. Using what you know about LCM, explain why 60 is a particularly useful number. (Hint: How many factors does 60 have? How does this help with dividing time and angles?)
